

EXHIBIT K

**IN THE UNITED STATES DISTRICT COURT
FOR THE DISTRICT OF COLORADO**

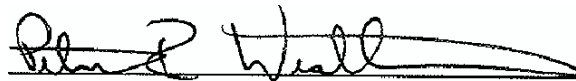
United States of America

v.

***CEMEX, Inc.
Lyons Cement Plant
Lyons, Colorado***

Civil Action No. 09-cv-00019-MSK-MEH
U.S. District Court for the District of Colorado

**EXPERT REBUTTAL REPORT
OF
PETER R. WESTLIN**

A handwritten signature in black ink, appearing to read "Peter R. Westlin", written over a horizontal line.

June 15, 2011

ON BEHALF OF THE UNITED STATES OF AMERICA

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Executive Summary

In my capacity as a technical expert, I reviewed the nitrogen oxides (NO_x) continuous emissions monitoring system (CEMS) reported data and other related information regarding the CEMEX Lyons Portland cement kiln located in Lyons, Colorado (Lyons facility) and the Expert Report submitted on behalf of CEMEX by John Hofmann¹ (Hofmann Report). The United States Department of Justice asked me to review and comment on whether the calculation of emissions in Section 3.2 of the Hofmann Report accurately characterizes the output of the CEMEX CEMS, with respect to correction of measured data for stack gas moisture content, as reported from the CEMS data acquisition system (DAS) for the period from 2000 through 2001.

My findings are summarized as follows:

- CEMEX's NO_x CEMS was subjected to periodic quality assurance and quality control audits (RATAs) during the period from 2000 through 2001. With one exception in March 2001, the RATAs showed that the CEMS met the relative accuracy requirements as specified in 40 CFR Part 60, Appendix B, Performance Specification 2, for data expressed in units of NO_x concentration and mass emissions rates (*i.e.*, ppmv and lb/hr).
- The DAS used with the CEMEX CEMS for most of the 2000 through 2001 period appropriately corrected for stack moisture content.

¹ Expert Report of John E. Hofmann, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado; John E. Hofmann, President and CEO of Trinity Consultants, Dallas, Texas; May 10, 2011.

INTRODUCTION

A. Information Required by Federal Rules of Civil Procedure

The following is a list of the items provided as required by the Federal Rules of Civil Procedure:

1. This report contains my opinions, conclusions, and the reasons therefore;
2. Appendix A contains a statement of my qualifications;
3. Appendix B lists the reports, data, and other information I considered in forming my opinions;
4. I have received no compensation for the preparation of this report or for my testimony other than my federal salary;
5. I have provided testimony within the preceding four years as an expert at trial or deposition.²

B. Areas of Requested Opinion

The U.S. Department of Justice (DOJ) asked me to provide opinions on the Hofmann Report conclusions with respect to the stack gas moisture adjustments in the calculation of emissions as determined by the CEMEX CEMS and the associated DAS for the period from 2000 through 2001.

In formulating my opinion on this subject, I relied on my more than 39 years of knowledge and experience gained while working for the U.S. Environmental Protection Agency (EPA) in various engineering and management positions in the Office of Air Quality Planning and Standards (OAQPS), Office for Air and Radiation. I relied particularly on the knowledge gained over those years in my roles developing and supporting regulatory activities related to emissions test method development and implementation, developing and supporting regulations that apply to CEMS applications, and developing and managing emissions data for regulatory and emissions factor development and application. Appendix A includes a more detailed description of my qualifications.

In formulating my opinions, I had access to all documents produced by the United States and CEMEX in this matter. I reviewed the documents provided to me and focused on those that pertained particularly to the operation and evaluation of the NO_x CEMS installed on the kiln at the Lyons facility and data management and calculation procedures of the CEMS. Among the documents to which I referred in my assessment are reports of compliance and performance tests and CEMS relative accuracy assessments, permit applications and supporting documents,

² I testified as an expert witness on September 10, 2009 in the case US v. ALCOA at TXU Sandow Development Company LP, Civil Action No. A-03-CA-222-SS.

descriptions and diagrams of the Lyons facility and the CEMS installation, quarterly CEMS audit reports, quarterly excess emissions reports, depositions of current and former CEMEX employees, and the Hofmann Report.

C. Experience

I began my service with EPA in 1971 as an environmental engineer in the Air Pollution Training Institute of OAQPS. Among other activities, I produced and provided a one-week training workshop to private industry and regulatory staff on the control of particulate matter. In 1972, I joined the Emissions Measurement Branch, where, for 19 years, I conducted field and laboratory studies of methods for measuring particulate and gaseous stack emissions components and used that information to prepare and publish stationary source emissions test methods and background documents for use in implementing federal and state air pollution regulations. Among those technical and regulatory development projects were long-term field evaluations of the performance of gaseous sulfur dioxide (SO₂) and NO_x CEMS. The purpose of these projects was to collect data and performance information that I then used to support and publish CEMS quality assurance procedures (*i.e.*, Procedure 1, Appendix F, 40 CFR Part 60) as well as the new source performance standard for fossil fuel-fired utilities, subpart Da.³ Relevant to the subject of this report, I also led the publication of the instrumental procedures for measuring gaseous emissions, Methods 3A and 6C of 40 CFR Part 60, Appendix A.

In 1991, I took a position as chief of the Test Support Section of the Emissions Standards and Engineering Division, directing as many as 19 engineers, chemists, and other staff in the development of new and revised test methods and new performance specifications for assessing CEMS, including those for gaseous organic pollutants. The CEMS performance specifications revised, developed, and published during this period include Performance Specifications 2, 7, and 8 of 40 CFR Part 60, Appendix B. Since 1994, when the office reorganized, I have been in the non-management position of senior engineer involved in regulatory support work.

I have also taken an active, leadership role in revamping the emissions factor program, particularly in the management of AP-42, assuring that the quality of data used in the program is known and that those emissions factors are applied appropriately. This work has included developing an approach to, and completing an assessment of, the statistical uncertainties associated with emissions factors and applying them for facility-specific purposes.

My work at EPA frequently requires the use of basic statistical procedures common to engineering practices. Although I am not a formally trained statistician, I have used statistical analyses in assessing emissions data to develop regulatory standards and procedures. I have also applied the statistically based procedures incorporated into EPA regulations in the conduct of numerous emissions testing and CEMS evaluation projects.

³ 40 CFR Part 60, Subpart Da, Standards of Performance for Electric Utility Steam Generating Units for Which Construction is Commenced After September 18, 1978, Section 60.47a; 40 CFR Part 60, Appendix F, Procedure 1, Quality Assurance Requirements for Gas Continuous Emission Monitoring Systems Used for Compliance Determination.

I continue in my role as senior engineer in advising rule developers and emissions measurements staff and management about the use of emissions test methods and CEMS in new and revised stationary source hazardous and criteria air pollutant control regulations.

II. CONTINUOUS EMISSIONS MONITORING SYSTEMS

A. Background

CEMS have been used to measure air pollution stack emissions in the United States for more than 35 years. EPA first required CEMS through regulations in the mid-1970's for purposes of monitoring the operation of SO₂, NO_x, and particulate control equipment used at fossil-fuel fired electric utilities. Source categories subject to gaseous CEMS applications in subsequent federal regulations include sulfuric and nitric acid plants, petroleum refineries, chemical plants, Kraft pulp mills, gas processing facilities, and incinerators. The data from CEMS are used to indicate good operation and maintenance and to determine compliance on a continuous basis with applicable emissions limitations.⁴

The term CEMS refers to the total system used to determine gas concentrations or emissions rates from pollutant sources. The major components, or subsystems, of a CEMS are the sample interface, pollutant and diluent analyzers, gas flow rate analyzer, and data recording and management apparatus (data acquisition system or DAS).⁵ The sample interface is used for sample acquisition, sample delivery, sample conditioning, and/or protection of the monitor from the effects of the stack effluent. The pollutant analyzer senses the pollutant gas (*e.g.*, NO_x) and generates an output proportional to the gas concentration. The diluent analyzer senses the component of gas (*i.e.*, CO₂ or O₂) used to correct for dilution or fuel combustion characteristics and generates an output proportional to the gas concentration. A gas flow rate analyzer usually detects gas velocity and generates a signal output proportional to the stack gas flow rate. The data recorder or data acquisition system (DAS) is used to provide a permanent record of the analyzers measurements. The DAS not only records analyzer data but also can manipulate signals according to integral algorithms and convert the resulting data to the required reporting units (*e.g.*, lb/hr) and averaging times. The DAS can also perform controlling functions for the analyzers, including calibrations and adjustments.

The most common type of gas concentration measurement component of a CEMS is the extractive system, in which a sample of gas is drawn or extracted from the stack or duct and transported through conditioning (*e.g.*, filtering, drying, and cooling) components to concentration measurement analyzers. The extraction usually occurs through a probe, a metal or other nonreactive material tube that is inserted into the gas stream and often heated to prevent

⁴ Westlin, Peter R., "A History and Status of CEMS Applications in USEPA Regulations," paper presented at the 1997 CEM Conference, Tedington, England, 1997.

⁵ 40 CFR Part 60, Appendix B, Performance Specification 2, Specifications and Test Procedures for SO₂ and NO_x Continuous Emission Monitoring Systems in Statutory Sources.

condensation. The sample often travels more than 100 feet through nonreactive tubing and sample conditioning components to the analyzers.

A critical CEMS function is to measure gaseous component concentrations. These components include pollutants (*e.g.*, SO₂, NO_x, CO, and organic compounds) and diluent gases (*e.g.*, O₂ and CO₂). Concentrations of diluent gases are used to correct pollutant concentrations to a standard condition (*e.g.*, 3 percent O₂) and to adjust measured data in calculating mass emissions rates. Analytical techniques for determining gaseous component concentrations include absorption spectroscopic, luminescence, electroanalytical, and paramagnetic methods. Measurement of multiple gaseous constituents often requires use of multiple analyzers. In many cases, a single extracted sample is shared through a distributive header with multiple analyzers for nearly simultaneous measurements.

Some emissions standards specify the continuous monitoring of pollutant mass emissions rates and thus require a measure of gas volumetric flow as part of the CEMS, in order to convert concentration data (*e.g.*, parts per million volume or ppmv) to mass emissions rate values (*e.g.*, pounds per hour or lb/hr). A CEMS that measures mass emissions rates is sometimes called a continuous emissions rate monitoring system, or CERMS. Such a system would include a flow rate measurement feature to convert pollutant concentration values to mass emissions rate values. There are a number of techniques available to measure volumetric flow rate in stacks or ducts. Direct measurement approaches, which are inherently *in-situ*, include differential pressure sensors (*e.g.*, Pitot devices, orifice and Venturi meters), vane anemometers, thermal sensors, and acoustic velocity measurements. Other gas flow measurement approaches include methods that correlate gas velocity or flow rate with measured operational parameters, such as electrical power and pressure data, used in conjunction with calibrated fan or blower flow curves. Less direct measurement methods include the use of steam flow rate and combustion stoichiometric data to calculate flow rates.

Industrial operators use all varieties of direct gas, indirect gas, and product flow monitoring devices in many process control and feedback operations. These data are used to help assure product quality and the efficient use of fuel and raw materials. The most numerous users of gas flow rate monitoring devices for regulatory compliance monitoring are sources complying with EPA's acid rain program.⁶ There are other local and state regulations and some federal rules involving mass emissions rate measurements (*e.g.*, lb/ton of product) for determining compliance. These types of regulations also require the use of flow rate monitoring devices. Most sources complying with the acid rain regulations rely on direct in-stack flow measurement devices such as ultrasonic (*i.e.*, acoustic) and differential pressure measurement techniques.

The determination of pollutant mass emissions rates involves the use of the pollutant concentration data and the gas flow rate data in a calculation or algorithm embedded in the DAS. As noted above, the DAS is a critical component of the CEMS that collects, records, and reports the measurement data in formats that are useful to the source operator and are required by the

⁶ 40 CFR Part 75.

applicable regulations. Signals from pollutant and diluent concentration analyzers, flow rate measurement devices, and other relevant detectors (*e.g.*, temperature) are manipulated mathematically, usually by a computer, to produce data in units of the applicable requirements, including mass emissions rate values.

B. Performance Specifications and Quality Assurance Assessments for CEMS

As noted above, EPA has enacted performance specifications (PS) designed to verify the reliability of a CEMS in measuring pollutant emissions so that the Agency and other environmental regulatory agencies may rely on CEMS data to ensure that air quality standards are being met throughout the country. Many stationary source emissions test methods, sometimes called reference test methods, and the PS are set forth in 40 CFR Part 60, Appendices A and B, respectively.

In 1975, EPA published Performance Specification 2 (PS 2), which is applicable to SO₂ and NO_x CEMS.⁷ PS 2 includes sample collection location requirements, data quality criteria, and, most critically, relative accuracy requirements. PS 2 contains only a few specific equipment design requirements, such as the range requirement of a data recorder scale and the ability to measure calibration drift.⁸ Instead, PS 2 and other PS contain procedures and acceptance criteria that are performance based and not specific to any particular measurement technology. That is, by focusing on accuracy and other data reliability requirements, the PS and associated regulations permit a source owner to apply any technology to establish and assess the CEMS performance in accordance with the PS. The PS prescribes testing procedures and calculations that a CEMS, as installed and operated at each subject facility, must undergo to demonstrate minimum performance requirements for accuracy and repeatability. Accuracy for the purposes of the PS is a measure of the difference between the CEMS measurements and those of the reference test method (see discussion of test methods below, at page 9) concurrently conducted. Repeatability is a measure of the CEMS's ability to respond consistently to a known input applied at regular intervals of time (*e.g.*, daily calibration gas injection).

The purpose of the PS is to provide a framework for assessing the ability of a CEMS to produce accurate and reliable data to demonstrate compliance with an applicable emissions limit or standard. Application of the PS requires the source owner to alert the appropriate regulatory authority of the field testing, conduct the relative accuracy testing and associated quality control (QC) protocols, and issue a report of the findings to the regulatory authority, usually in accordance with a date specified in the applicable rule or operating permit.⁹ The regulatory authority then approves or disapproves the report. If the regulatory authority approves the findings, the CEMS is considered to be "certified," meaning that the source owner may rely on the CEMS data for demonstrating compliance with applicable emission standards. To maintain the CEMS certification, the general provisions, an applicable rule, or the permit will require ongoing quality assurance and quality control (QA/QC) activities, including conducting daily

⁷ 40 CFR Part 60, Appendix B, Performance Specification 2.

⁸ 40 CFR Part 60, Appendix B, Performance Specification 2, Section 6.0.

⁹ 40 CFR Part 60.7(a)(5).

calibrations, quarterly calibration gas audits, and periodic PS demonstrations, discussed further below at pages 10 – 11.¹⁰

The relative accuracy requirement is the most important element of the certification of a CEMS. The relative accuracy of a CEMS is determined through a series of comparisons checks, during which two measurements are compared to each other: (1) the recorded CEMS data and (2) the results of the contemporaneous (manual or instrumental) reference test method. PS 2 specifies that a relative accuracy test consists of at least nine comparison checks or runs.¹¹ The appropriate reference method is specified in the applicable regulation. For SO₂ and NO_x CEMS, the acceptable relative accuracy (RA) result is 20 percent, as specified in PS 2,¹² and is expressed in units of the applicable standard (*e.g.*, lb/hr of NO_x). The RA is not simply the average difference between the two measurements but is expressed as the average difference between the reference test method data and the CEMS data plus a confidence coefficient derived from the variability of the comparisons.¹³ In this calculation, the RA accounts for both the magnitude of the differences between the two measurements and the variability of those measurement differences. The relative accuracy principle is essential to assessing the quality of CEMS data for compliance determination purposes and is basic to all CEMS applications for implementing EPA regulations.

The general provisions of the New Source Performance Standards include requirements applicable to installation and operation of a CEMS.¹⁴ These requirements address installation requirements, including location, for assuring that the CEMS provides measurements representative of emissions and process parameters for the affected facility. These requirements also refer to applicable QA/QC requirements that apply to Appendices B and F for use in certifying and operating a CEMS that is used to determine compliance with applicable requirements.

EPA has published numerous detailed test methods and specified their use in conducting the CEMS relative accuracy testing as well as for demonstrating initial compliance with air regulations for the affected facility. The principles behind many of the test methods have been derived from laboratory procedures and adapted for use in the field. Most of the methods EPA has published require that samples be extracted from exhaust stacks, collected in an appropriate medium, and then stored and shipped to a laboratory for analysis using physical and chemical laboratory techniques. Since the late 1970's, EPA has published methods based on instrumental analytical techniques using analytical technologies similar to those described above for extractive CEMS analyzers. These instrumental methods measure pollutant concentrations on site by drawing representative samples through lines and conditioning systems to direct interface with the analyzer or in aliquots of integrated samples injected into an on-site analyzer immediately after sample collection.

¹⁰ 40 CFR 60.13(a)-(j). *See also, e.g.*, 40 CFR Part 60, Appendix F, Procedure 1; 40 CFR 60.47a.

¹¹ 40 CFR Part 60, Appendix B, Performance Specification 2, Section 8.4.4.

¹² *Id.* at Section 13.2.

¹³ *Id.* at Section 3.9 and Equation 2-6.

¹⁴ 40 CFR 60.13(a), (d), and (f).

The following test methods, set forth in 40 CFR Part 60, Appendix A, are relevant to the subject matters of this report:

- o Method 1, Sample and Velocity Traverses for Stationary Sources, defines physical location of sampling ports and points in the stack or duct for collecting representative samples for analysis or for measuring gas velocity;
- o Method 2, Determination of Stack Gas Velocity and Volumetric Flow Rate (Type S Pitot Tube), describes the techniques and data quality assurance requirements for measuring and calculating stack gas flow rates;
- o Method 3A, Determination of Oxygen and Carbon Dioxide Concentrations in Emissions from Stationary Sources (Instrumental Analyzer Procedure), prescribes the use of instrumental measurement technologies and procedures for measuring diluent gas concentrations and calculating pollutant concentrations corrected for dilution;
- o Method 3B, Gas Analysis for the Determination of Emission Rate Correction Factor or Excess Air, describes wet chemistry equipment and procedures for measuring diluent gas concentrations and calculating pollutant concentrations corrected for dilution;
- o Method 4, Determination of Moisture Content in Stack Gases, specifies equipment and procedures for measuring stack gas moisture content;
- o Method 7E, Determination of Nitrogen Oxide Emissions from Stationary Sources (Instrumental Analyzer Procedure); and
- o Method 19, Determination of Sulfur Dioxide Removal Efficiency and Particulate Matter, Sulfur Dioxide, and Nitrogen Oxide Emission Rates.

To ensure the ongoing quality of CEMS data used for determining compliance and potentially establishing violations and associated fines, EPA has published additional CEMS data QA procedures in 40 CFR Part 60, Appendix F, effective 1979.¹⁵ Procedure 1 of Appendix F addresses two distinct and equally important functions. One function is the periodic assessment of the quality of the CEMS data by estimating accuracy; the other function is the control and improvement of the quality of the CEMS data by implementing quality control policies and corrective actions.¹⁶ These two functions form a control loop: when the assessment function indicates that the data quality is inadequate, the control effort must be increased until the data quality is acceptable.

In order to provide uniformity in the assessment and reporting of data quality, EPA QA procedure specifies the assessment methods for CEMS response drift and accuracy, including

¹⁵ 40 CFR Part 60, Appendix F, Procedure 1.

¹⁶ 40 CFR Part 60, Appendix F, Procedure 1, Section 1.2.

periodic relative accuracy testing. CEMS response drift is a measure of the ability of the CEMS to respond with the same output periodically (*e.g.*, daily) to the same reference value, usually a cylinder gas of known concentration. Among other procedures and criteria, Procedure 1 also prescribes a quarterly cylinder gas audit (CGA) that challenges and assesses the abilities and calibration of the separate gaseous component analyzers against the known standards of a certified calibration, or Protocol 1, cylinder gas.¹⁷

Once a CEMS is qualified according to the applicable PS and is subjected to and continues to pass the ongoing QA/QC requirements, to determine compliance, the applicable rules require that data from a CEMS so assessed and found acceptable be reported without adjustment. That is, neither the PS 2 nor Appendix F of Part 60 provides a procedure to allow a source to adjust its CEMS emissions data when reporting the data to demonstrate compliance with applicable emissions limits. The structure and application of PS 2 and Appendix F presume that data from a qualified CEMS are representative of site-specific emissions. Therefore, for most applications, EPA requires that sources submit CEMS emissions data as recorded, without adjustment, so that the data may be used to assess compliance with applicable emissions limits or standards.¹⁸

III. CEMEX KILN NO_x CEMS

A. Measurement Equipment and Operation

The 1993 Lyons kiln construction permit required that CEMEX install and operate a CEMS for measuring NO_x mass emissions rate reported in pounds of NO_x per hour (lb/hr) and tons of NO_x per year (t/y).¹⁹ The values reported from the CEMS were of the kiln NO_x emissions only and did not include the NO_x emissions from the dryer (*i.e.*, the dryer had a separate fuel supply and a separate exhaust point and was subject to a separate construction permit). By August 1995, CEMEX had installed a cold, dry extractive sampling and analysis system for measuring concentrations of NO_x in parts per million by volume dry (ppmvd) and a stack gas flow rate measurement component for measuring NO_x mass emissions rates (*e.g.*, lb/hr).²⁰ A cold, dry extractive sampling technique is one that draws a representative portion of gas from the stack and routes it through a heated line to a moisture removal apparatus that also cools the gas before conveying a fraction of the gas to the analyzer. The analyzer in this case was a Perkin-Elmer MCS-100 model that operated on a nondispersive infrared absorption principle. This analytical device also measured concentrations of carbon dioxide (CO₂), carbon monoxide (CO), and sulfur dioxide (SO₂) in ppmvd concentrations.

¹⁷ EPA Traceability Protocol for Assay and Certification of Gaseous Calibration Standards, EPA, NERL, September 1979.

¹⁸ 40 CFR 60.7(c).

¹⁹ Colorado Department of Public Health and Environment (CDPHE) Construction Permit, Permit No: 12BO444-2, December 17, 1993, US00006499 at 6525-6526.

²⁰ Carnot, Test Report for the 1995 Relative Accuracy Test Audit at the Southwest Portland Cement Company, October 5, 1995, CEMEX_NEIC_00008722, beginning at 8725.

The stack gas flow rate measurement portion of the CEMS was a United Sciences, Inc., Ultraflow 100 in-stack device operating with a time-of-flight, or ultrasonic, principle to produce data in standard cubic feet per minute at wet stack conditions (scfmw).^{21,22} Addition of the flow rate measurement allowed the CEMS to calculate and report NO_x mass emissions rate values as lb/hr and to calculate annual emissions. The CEMS with the cold extractive sampling and MCS-100 analyzer with the Ultraflow 100 gas flow measurement device was the configuration used to measure the NO_x emissions rates during 2000-2001, the period that is the subject of this report. CEMEX also had CEMS for measuring opacity, total gaseous organic emissions, and oxygen.²³ These other stack gas constituent measurements did not figure into the measurements necessary for determining the NO_x mass emissions rate.

B. Data Management and Calculations

Sometime in November 1999, CEMEX installed and operated a DAS produced by WTC Engineering.²⁴ The WTC DAS reported hourly concentration, flow rate, and mass emissions rate values, and the system calculated and reported a mass rate (*i.e.*, lbs NO_x /hour) value. The WTC DAS also identified some periods for which data were not used to calculate the total mass of NO_x emissions, albeit with different codes and flags. In calculating the pollutant mass emissions rates, the WTC DAS appeared to sum the hourly mass values, those without any data flags, to produce a monthly mass emissions value.

As described above, the NO_x concentrations were measured on a dry basis and the stack gas flow rate was measured on a wet basis. The different bases for the measurements meant that the DAS calculation processor must convert the measured values to the same moisture basis to calculate the mass emissions rate. The commonly understood approach is for the DAS to convert the stack flow rate measurement (*i.e.*, wet basis) to a dry basis using a value representing the stack moisture content. The equation that includes a correction for stack gas moisture in calculating NO_x emissions in units of lbs/hour is:

$$NO_x \text{ emissions } \left(\frac{lbs}{hr} \right) = NO_x \text{ (ppm)} * Q \left(\frac{ft^3}{hr} \right) * 1.194 * 10^{-7} \left(\frac{lb}{ppm * ft^3} \right) * \left(1 - \frac{\%H_2O}{100} \right)$$

In this equation:

- NO_x(ppm) = concentration of NO_x (nitrogen oxide and nitrogen dioxide reported as NO₂) in ppm by volume dry;
- Q = Stack flow rate in wet cubic feet per hour (ft³/hr);

²¹ Deposition of Monica Sowders, January 5, 2001, TR at 117:15 – 118:2.

²² Ultraflow 150 Gas Flow and Temperature Monitor Operations Manual, June 2007, CEMEX_NEIC_00012674, beginning at 12675.

²³ Air Pollution Testing, Inc., Relative Accuracy Test Audit for Southdown Inc.: Lyons, Colorado Facility, Kiln Stack CEMS, January 20, 2000, CEMEX_NEIC_00009667, beginning at 9668.

²⁴ Deposition of Gregory Wright, November 9, 2010, TR at 15:6 - 14 (“November of 1999 in fact.”). *See also*, Deposition of Monica Sowders, January 5, 2011, TR at 86:10 - 14 and 86:19 - 22; Deposition of Chuck Kendrick, January 11, 2011, TR at 8:20 - 24.

- 1.194×10^{-7} = Constant of conversion of NO_x (ppm) to lb/dry ft^3 ;²⁵ and
- % H_2O = Percent by volume water vapor in the stack gas.

The CEMEX kiln CEMS was typical of most process emissions monitoring in that it did not directly measure the stack gas moisture content. Instead, as do many such CEMS, the CEMEX CEMS data file²⁶ shows that prior to 2000, the CEMS DAS computer code applied a constant, 19 percent moisture, for the NO_x concentrations and stack flow rate conversion to a mass emissions rate in lb/hr. This constant or default moisture concentration for the calculations was apparently based on the results of emissions testing conducted for the CEMS relative accuracy projects.²⁷

The WTC Variables List²⁸ provided by Chuck Kendrick of WTC indicates that the default stack moisture content value remained 19 percent (for the emissions rate calculations) from 1999 through October 2001 while the WTC DAS was in use.

C. CEMS RATA Reports

The NO_x CEMS underwent periodic audit testing in accordance with Performance Specification 2, 40 CFR Part 60, Appendix B (PS 2), also called the relative accuracy test audit, or RATA.

January 2000. Air Pollution Testing, Inc., an emissions testing and consulting firm, conducted the RATA for the CEMEX kiln NO_x CEMS in January 2000.²⁹ The results of this RATA were that the unit satisfied the PS 2 relative accuracy requirements for NO_x concentration, ppmvd, measurements (relative accuracy of 6.9 percent) and for gas flow rates (relative accuracy of 9.5 percent). The report did not include the relative accuracy results for units of mass emissions rates, lb/hr. Using the emissions concentration and flow values reported for this RATA and a default stack gas moisture value of 19 percent, I calculated the mass emissions rates, lb/hr, for each run and the relative accuracy results in units of mass emissions rate. (see Appendix C) I found the RA result to be 16.4 percent. This result also satisfied the PS 2 relative accuracy requirement. I conclude that the CEMS was operating within the necessary RATA requirements in January 2000 and the DAS was adequately correcting for stack gas moisture content.

²⁵ EPA, Method 19—Determination of Sulfur Dioxide Removal Sufficiency and Particulate Matter, Sulfur Dioxide and Nitrogen Oxide Emission Rates, Table 19-1 at *22, <http://epa.gov/ttn/emc/promgate/m-19.pdf>, last visited June 7, 2011.

²⁶ Lyons Criteria Pollutant Emission Data, 1994, CEMEX_NEIC_00008571, beginning at 8572.

²⁷ Defendant CEMEX, Inc. Responses to United States' First Set of Interrogatories, Interrogatory Response 6, beginning at page 19.

²⁸ Addendum to Deposition of Chuck Kendrick, January 11, 2011, Bin Definitions for Southdown: Lyons.

²⁹ Air Pollution Testing, Inc., Relative Accuracy Test Audit for Southdown Inc.: Lyons, Colorado Facility, Kiln Stack CEMS, January 20, 2000, CEMEX_NEIC_00009667, beginning at 9668.

March 2001. Air Pollution Testing, Inc., also conducted the RATA for the CEMEX kiln NO_x CEMS in March 2001.³⁰ The results of this RATA were that the unit satisfied the PS 2 relative accuracy requirements for NO_x concentration, ppmvd, measurements (relative accuracy of 15.7 percent) and for stack gas flow rate (relative accuracy of 19.7 percent). The report did not include the relative accuracy results for units of mass emissions rates, lb/hr, for this RATA. A letter from CEMEX to CDPHE immediately after submitting the test report indicated a significant problem with the flow-rate monitoring component of the CEMEX CEMS and “an error in the nitrogen oxides (NO_x) mass emission rate CERMS calculation....”³¹

I could not determine what the calculation or measurement error was. Using the reported CEMS gas flow rate values and the NO_x concentration data and a default stack gas moisture value of 19 percent, I was able to reproduce the CEMEX CEMS NO_x mass emissions rate values listed in the RATA report. I also calculated the CEMS relative accuracy for the data in units of mass emissions rate, lb/hr, and found the result to be 39.5 percent (see Appendix C). As the report states, this result does not satisfy the PS 2 relative accuracy requirement. I conclude that the NO_x CEMS was operating within the necessary RATA requirements for NO_x concentration measurements but not for mass emissions rates in March 2001. At this point, the DAS was adequately correcting for stack gas moisture content, but, for other reasons, the CEMS was not producing reliable NO_x mass emissions rate data.

June 2001. Air Pollution Testing, Inc., also conducted the RATA for the CEMEX kiln NO_x CEMS in June 2001.³² The results of this RATA were that the unit satisfied the PS 2 relative accuracy requirements for NO_x concentration, ppmvd, measurements (relative accuracy of 0.6 percent) and in units of mass emissions rates, lb/hr, (relative accuracy of 10.3 percent). I conclude that the CEMS was operating within the necessary RATA requirements in June 2001 and producing reliable emissions data.

IV. HOFMANN REPORT AND MOISTURE CORRECTIONS

A. Hofmann Report Summary and Conclusions

In section 3.2 of the Hofmann Report³³, Mr. Hofmann describes the CEMEX CEMS operation in generally same manner as I have described above. He similarly discusses the background of the need for the moisture correction calculation and the historical approach applied to the CEMEX CEMS in calculating the emissions data.³⁴ In his report, Mr. Hofmann concludes that the “The WTC software was not ‘moisture correcting’ the NO_x concentrations.”³⁵

³⁰ Air Pollution Testing, Inc., Source Emissions Test Report, CEMEX, Inc., Kiln Stack CEMS, March 6, 2001, CEMEX_NEIC_00009845, beginning at 9846.

³¹ Letter from Paul Ottenstein, CEMEX Technical Director, to Long Nguyen, Air Pollution Control Division (APCD) Permit Engineer, Re: Southdown Lyons Cement Plant, May 7, 2001, CMXLYO00018625 at 18628.

³² Air Pollution Testing, Inc., Source Emissions Test Report, CEMEX, Inc., Kiln Stack CEMS, June 21, 2001, CEMEX_NEIC_00009955, beginning at 9956.

³³ Hofmann Report, Section 3.2, p. 3-2 through 3-4.

³⁴ Hofmann Report, p. 3-2 to 3-4.

³⁵ Hofmann Report, p. 3-3.

What Mr. Hofmann means by this is that the CEMEX CEMS DAS was incorrectly calculating the NO_x mass emissions rates by not correcting the stack gas flow rate data to dry conditions. Mr. Hofmann cites the depositions of Monica Sowders and Chuck Kendrick^{36,37} to support this conclusion. He notes that he did not review any raw CEMS data, RATA results, or the DAS programming logic or correction factors.³⁸

Based on his conclusion about the DAS programming, Mr. Hofmann indicates that CEMEX is reasonable and correct in assuming that the reported NO_x mass emissions data for 2000-2001 are not corrected for moisture. More to the point, he concludes that NO_x mass emissions rate values reported for the entire 2000-2001 period should be adjusted to account for the stack gas moisture content in order to address this calculation error.³⁹

B. Rebuttal Discussion

In my review of the available CEMEX CEMS-related documentation, I found information leading to a conclusion significantly different than that Mr. Hofmann reached. A September 2007 internal memo from Matthew Parks to Paul Carr, Unit Supervisor, CDPHE-APCD, described an error in the WTC DAS relative to the stack moisture correction calculation.⁴⁰ In that memo, Mr. Parks asserted that the CEMEX CEMS DAS was correctly accounting for stack moisture prior to 2002. I found additional confirmation that the CEMS DAS was appropriately accounting for moisture over this period in the CDPHE Field Inspection Report⁴¹ prepared in November 2007 and a Notice of Violation CDPHE issued to CEMEX in 2008.⁴² These documents state that the CEMS DAS “was correcting for stack moisture content as described in the Continuous Monitoring System Quality Assurance Plan...”⁴³ through some part of 2001.

Chuck Kendrick confirmed in his deposition pursuant to Fed. R. Civ. P. 30(b)(6) on behalf of WTC that the WTC DAS and the associated software installed at the CEMEX facility in late 1999 incorporated the same calculations, including moisture adjustments, as the Odessa DAS, which was in place prior to that time.⁴⁴ In his deposition, Mr. Kendrick indicated that he could confirm that the CEMEX DAS was appropriately correcting for stack moisture content through October 2001.

³⁶ Deposition of Monica Sowders, January 5, 2001.

³⁷ Deposition of Chuck Kendrick, January 11, 2011.

³⁸ Hofmann Report, p. 3-4.

³⁹ Hofmann Report, p. 3-4.

⁴⁰ Memorandum from Matthew Parks to Paul Carr, Unit Supervisor, CDPHE-APCD, Re: CEMEX CEMS, September 10, 2007, CEMEX_NEIC_00008623 at 8624 - 8626.

⁴¹ APCD Field Inspection Report, November 11, 2007, US00057313.

⁴² CDPHE Notice of Violation, Case No. 2008-037, CMXLYO00018689.

⁴³ Continuous Monitoring System, Quality Assurance Program, CEMEX Lyons, Colorado, last revised September 28, 2006, CMXLYO00019125.

⁴⁴ Deposition of Chuck Kendrick, January 11, 2011, TR at 34:12 - 23.

In its separate summary emissions report of the annual emissions for 1994 through 2007, CEMEX indicated⁴⁵ that the WTC DAS was appropriately correcting for stack moisture content through at least 2000. Also, as noted above, I used data provided in the testing contractor's report of the RATA conducted in January 2000 and June 2001 and confirmed that the CEMS was appropriately correcting for moisture at those times.^{46, 47} I carried out these calculations with an assumed 19 percent stack moisture content.

Based on the depositions and other documentation cited above and my own calculations, I conclude that the CEMEX CEMS DAS was appropriately correcting for stack moisture in the calculation of NO_x mass emissions rate from the start of 2000, clearly through June 2001, and probably through October 2001. That means that for at least the first 18 months (and probably the first 22 months) of the 2000-2001 period, the NO_x mass emissions data produced by the CEMEX CEMS was appropriately correcting for stack gas moisture content and any further adjustments for moisture would be unnecessary and improper.

V. SUMMARY AND CONCLUSIONS

A. CEMS Relative Accuracy Test Audits

The CEMEX CEMS RATA results spanning 2000 through 2001 generally indicated that the CEMS was operating within the performance criteria as specified in PS 2 and Appendix F, Procedure 1, with respect to NO_x concentration and mass emissions rate data. I conclude from this review of the relative accuracy data for both NO_x concentration and mass emissions rate that the CEMS produced somewhat variable but generally reliable NO_x mass emissions from the beginning of 2000 through at least June 2001 with the exception of March 2001.

B. Correcting for Stack Gas Moisture Content

For the reasons cited in section IV. B. above, I conclude that the CEMEX CEMS DAS was correctly adjusting for stack moisture content for the data reported for 2000, through at least June of 2001, and probably until October of that year. This contrasts with the opinion offered by John Hofmann, who concludes that CEMEX should be corrected to adjust the reported NO_x mass emissions data downward for the years 2000 and 2001, to account for stack gas moisture in calculating NO_x mass emissions rates. Mr. Hofmann is incorrect when he states that, "it is reasonable for CEMEX to correct the previously reported NO_x emissions" because the CEMEX CEMS DAS was already correcting for stack gas moisture content during most of this period. (Hofmann 3-4) It is my opinion that CEMEX would be incorrect in adjusting the reported CEMS data for stack moisture content in calculating annual emissions for 2000 and much of 2001.

⁴⁵ CEMEX Lyons Historic Lyons NO_x Emissions Trends, CMXLYO00131241 at 131242.

⁴⁶ Air Pollution Testing, Inc., Source Emissions Test Report, CEMEX, Inc., Kiln Stack CEMS, June 21, 2001, CEMEX_NEIC_00009955, beginning at 9956.

⁴⁷ RATA Analysis Spreadsheet, Exhibit 1, filename: 1995_2001 RATA 021611.xlsx, tabs: Jan 2000 RATA, Mar 2001 RATA, Jun 2001 RATA.

Appendix A

Statement of Qualifications

Peter R. Westlin
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US Environmental Protection Agency
Research Triangle Park, NC 27711
Phone: (919) 541-1058

Current position: Senior Environmental Engineer, Monitoring Policy Group, Technical Support Division, Office Air Quality Planning and Standards, US Environmental Protection Agency

Position Description: Serves in the Measurement Policy Group (MPG) within the Sector Policies and Programs Division (SPPD) and fulfills the role as senior advisor and national expert on stationary source air emissions measurement, monitoring, and other approaches for quantifying air emissions from stationary sources. I work in conjunction within OAQPS and other EPA offices, other regulatory organizations, and industry in developing methods of identifying and resolving emissions quantification issues on the review, analysis, coordination and implementation of proposed, new, or revised source emissions measurement methods, monitoring approaches, regulations, application of emissions factors, and emissions quantification guidance documents.

Education:

Purdue University	MSE	Environmental Engineering, 1971
Purdue University	BSME	Mechanical Engineering, 1970

Previous Employment:

04/91 – 11/94	Section Chief, Emissions Measurement Branch, OAQPS
06/72 – 04/91	Environmental Engineer, Emissions Measurement Branch, Emissions Standards and Engineering Division, OAQPS
11/71 - 06/72	Air Pollution Training Institute, OAQPS

Publications within past 15 years

Westlin, Peter R., “The Status of Source Emissions Monitoring in the United States Air Program and Concepts for Continued Improvement,” paper presented at the 2006 CEM Conference, Paris, France, 2006.

Westlin, Peter R. “Summary of USEPA Instrumental Test Methods Used for Compliance Determination and CEMS Validation,” paper presented at the 1998 CEM Conference Tedington, England, 1998.

Westlin, Peter R., “A History and Status of CEMS Applications in USEPA Regulations,” paper presented at the 1997 CEM Conference, Tedington, England, 1997.

Federal test methods, CEMS performance specifications, and other regulatory support

- o 40 CFR Part 64, Compliance Assurance Monitoring Rule and associated technical guidance materials;
- o 40 CFR Part 60, Methods 3A, 6C, 25A, and 25B, Determination of Oxygen, Carbon Dioxide, Sulfur Dioxide, and Gaseous Organic Concentrations in Emissions from Stationary Sources (instrumental analyzer procedures);
- o Method 5D, Determination of Particulate Emissions from Positive Pressure Fabric Filters;
- o Method 19, Determination of Sulfur Dioxide Removal Efficiency and Particulate, Nitrogen Oxides Emissions Rates (using fossil fuel and organic waste analyses data);
- o Performance Specification 2, Specifications and Test Procedures for SO₂ and NO_x Continuous Emissions Monitoring Systems for Stationary Sources; and
- o Appendix F, Procedure 1, Quality Assurance Requirements for Gaseous Continuous Emissions Monitoring Systems Used for Compliance Determinations.
- o Recent significant technical and policy application contributions to regulations:
 - o Implementation of the PM_{fine} Ambient Air Quality Standard
 - o New Source Performance Standards and National Emissions Standards for Hazardous Air Pollutants
 - Fossil Fuel-Fired Utilities and Industrial Boilers
 - Portland Cement Plants
 - Commercial, Industrial Solid Waste Incinerators
 - Sewage Sludge Incinerators
 - Reciprocating Internal Combustion Engines
 - Nitric Acid Plants
 - Primary and Secondary Aluminum Production
 - Ferroalloy and Steel Production
 - Phosphate Rock and Phosphate Fertilizer Production

Appendix B

Documents Considered

Reports, data, and other information I considered in forming my opinions.

1. 40 CFR 60.13, Monitoring Requirements.
2. 40 CFR 60.47a, Emission Monitoring.
3. 40 CFR 60.7, Notification and Record Keeping.
4. 40 CFR Part 60, Appendix A, Method 1, Sample and velocity traverses for stationary sources.
5. 40 CFR Part 60, Appendix A, Method 2, Determination of Stack Gas Velocity and Volumetric Flow Rate (Type S Pitot Tube), describes the techniques and data quality assurance requirements for measuring and calculating stack gas flow rates.
6. 40 CFR Part 60, Appendix A, Method 3A—Determination of Oxygen and Carbon Dioxide Concentrations in Emissions From Stationary Sources (Instrumental Analyzer Procedure).
7. 40 CFR Part 60, Appendix A, Method 3B—Gas Analysis for the Determination of Emission Rate Correction Factor or Excess Air.
8. 40 CFR Part 60, Appendix A, Method 4, Determination of Moisture Content in Stack Gases (Specifies Equipment and Procedures for Measuring Stack Gas Moisture Content).
9. 40 CFR Part 60, Appendix A, Method 7E, Determination of Nitrogen Oxide Emissions from Stationary Sources (Instrumental Analyzer Procedure).
10. 40 CFR Part 60, Appendix A, Method 19—Determination of Sulfur Dioxide Removal Sufficiency and Particulate Matter, Sulfur Dioxide and Nitrogen Oxide Emission Rates, <http://epa.gov/ttn/emc/promgate/m-19.pdf>.
11. 40 CFR Part 60, Appendix B, Performance Specification 2 - Specifications And Test Procedures For SO₂ And NO_x Continuous Emission Monitoring Systems In Stationary Sources, <http://www.epa.gov/ttn/emc/perfspec/ps-2.pdf>.
12. 40 CFR Part 60, Appendix F, Procedure 1 - Quality Assurance Requirements For Gas Continuous Emission Monitoring Systems Used For Compliance Determination, <http://www.epa.gov/ttn/emc/perfspec/appendixf.pdf>.
13. 40 CFR Part 60, General Provisions, Monitoring Requirements, Sections 60.13(a)-(j).

14. 40 CFR Part 60, General Provisions, Notification and Recordkeeping, Section 60.7(a)(5).
15. 40 CFR Part 60, Subpart Da, Standards of Performance for Electric Utility Steam Generating Units for Which Construction is Commenced After September 18, 1978.
16. 40 CFR Part 63, Subpart S, National Emissions Standards for Hazardous Air Pollutants for Pulp and Paper Industry.
17. 40 CFR Part 75, Appendix A, Specifications and Test Procedures.
18. 40 CFR Part 75, Appendix F, Conversion Procedures.
19. 40 CFR Part 75, Emissions Monitoring (Acid Rain Regulations).
20. Air Pollution Testing, Inc., Relative Accuracy Test Audit for Southdown Inc.: Lyons, Colorado Facility, Kiln Stack CEMS, January 20, 2000, CEMEX_NEIC_00009667.
21. Air Pollution Testing, Inc., Source Emissions Test Report, CEMEX, Inc., Kiln Stack CEMS, March 6, 2001, CEMEX_NEIC_00009845.
22. Air Pollution Testing, Inc., Source Emissions Test Report, CEMEX, Inc., Kiln Stack CEMS, June 21, 2001, CEMEX_NEIC_00009955.
23. Boulder County Health Department, Air Inspection Report, September 5, 1995, CMXLYO000606557.
24. Carnot, Test Report for the 1995 Relative Accuracy Test Audit at the Southwest Portland Cement Company, October 5, 1995, CEMEX_NEIC_00008722, beginning at 8725.
25. Colorado Department of Public Health and Environment, Air Pollution Control Division, Field Inspection Report, November 11, 2007, US00057313.
26. Colorado Department of Public Health and Environment, Construction Permit, December 17, 1993, US00006499.
27. Colorado Department of Public Health and Environment, Notice of Violation, Case No. 2008-037, CMXLYO00018689.
28. Colorado Department of Public Health and Environment, Title V Operating Permit, Issued February 1, 2000, CMXLYO00013091.
 - a. Revised April 17, 2001, CMXLYO00013204;
 - b. Revised April 23, 2001, CMXLYO00013315;
 - c. Revised January 9, 2002, CMXLYO00013539;

- d. Revised March 26, 2002, CMXLYO000132427;
 - e. Revised October 8, 2002, CEMEX_NEIC_00000518;
 - f. Renewed March 1, 2008, CMXLYO00117950.
- 29. Continuous Monitoring System, Quality Assurance Program, CEMEX Lyons, Colorado, last revised September 28, 2006, CMXLYO00019125.
 - 30. CEMEX Lyons Historic Lyons NO_x Emissions Trends, CMXLYO00131241.
 - 31. Deposition of Chuck Kendrick, January 11, 2011.
 - 32. Deposition of Gregory Wright, November 9, 2010.
 - 33. Deposition of Monica Sowders, January 5, 2011.
 - 34. EPA, Emission Factor Documentation for AP-42, Section 11.6, Portland Cement Manufacturing, May 18, 1994, <http://www.epa.gov/ttn/chief/ap42/ch11/bgdocs/b11s06.pdf>.
 - 35. EPA, Emission Factors & AP42, Compilation of Air Pollutant Emission Factors, Volume 1: Stationary Point and Area Sources, <http://www.epa.gov/ttn/chief/ap42/index.html>.
 - 36. EPA, Memorandum: Performance Test Calculation Guidelines, June 6, 1990, <http://www.epa.gov/ttn/emc/informd/tid-024.pdf>.
 - 37. EPA, Office of Air Quality Planning and Standards, Emissions Factor Uncertainty Assessment, October 2007, <http://www.epa.gov/ttn/chief/efpac/uncertainty.html>.
 - 38. EPA, Traceability Protocol for Assay and Certification of Gaseous Calibration Standards, NERL, September 1979.
 - 39. Expert Report of John E. Hofmann, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado; John E. Hofmann, President and CEO of Trinity Consultants, Dallas, Texas; May 10, 2011.
 - 40. H2NS Environmental, About H2NS Environmental (including historic information about Odessa DSM data loggers), <http://www.h2ns.com/about.html>.
 - 41. Letter from Keith Baugues, McVehil-Monnett Senior Environmental Engineer, to John Lohr, CEMEX Plant Manager, Re: Update on Calculating Emissions: MMA Project Number 786-98, March 10, 1998, US00027788.
 - 42. Letter from Paul Ottenstein, CEMEX Technical Director, to Long Nguyen, APCD Permit Engineer, Re: Southdown Lyons Cement Plant, May 7, 2001, CMXLYO00018625.

43. Lyons Criteria Pollutant Emission Data, 1994, CEMEX_NEIC_00008571, beginning at 8572.
44. Lyons Plant Kiln Down Time for 2000, CMXLYO00153606.
45. Memorandum from Matthew Parks, Air Pollution Testing, Inc., Review/Technical Writer, to Paul Carr, APCD Unit Supervisor, Re: CEMEX CEMS, September 10, 2007, CEMEX_NEIC_00008623.
46. SEC 114 Request (Disc), CEMEX_NEIC_00008436.
47. Southdown, Inc., 2000 Month Emissions, October 28, 2003, CMXLYO00348833.
48. Technical Review Document for Title V Operating Permit 95OPBO082, September 1999, US00029363.
49. Ultraflow 150 Gas Flow and Temperature Monitor Operations Manual, CEMEX_NEIC_00012674.
50. Untitled Table, October 1, 1999, CMXLYO00206613.
51. Westlin, Peter R., "A History and Status of CEMS Applications in USEPA Regulations," paper presented at the 1997 CEM Conference, Tedington, England, 1997.

Appendix C

CEMS Relative Accuracy Example Calculations

Equations used in calculating CEMS emissions and relative accuracy in accordance with Performance Specification 2, 40 CFR part 60, Appendix B.

Moisture Correction

$$Q_d = Q_w (1 - B_{ws})$$

Where:

Q_d = stack gas flow rate corrected to dry basis (dscfm)

Q_w = stack gas flow rate as measured on wet basis (scfm)

B_{ws} = moisture content of stack gas, volume fraction

Arithmetic Mean

$$\bar{d} = \frac{1}{n} \sum_{i=1}^n d_i$$

Where:

n = Number of data points

$\sum_{i=1}^n d_i$ = Algebraic summation of the individual differences d_i

Standard Deviation

$$S_d = \sqrt{\frac{\sum_{i=1}^n d_i^2 - \frac{[\sum_{i=1}^n d_i]^2}{n}}{n - 1}}$$

Confidence Coefficient

$$CC = t_{Q, P, n} \frac{S_d}{\sqrt{n}}$$

Relative Accuracy

$$RA = \frac{[|\bar{d}| + |CC|]}{\overline{RM}} * 100$$

Where:

$|\bar{d}|$ = Absolute value of the mean differences

$|CC|$ = Absolute value of the confidence coefficient

$\overline{RM}$ = Average RM value. In cases where the average emissions for the test are less than 50 percent of the applicable standard, substitute the emission standard value in the denominator of the equation in place of $\overline{RM}$. In all other cases, use $\overline{RM}$.

Jan-00

	Field reference Method Data			CEMS Data			
Run	NO _x (ppmvd)	Q _d (dscfm)	NO _x (lb/hr)	NO _x (ppmvd)	Q _w	Q _d (dscfm)	NO _x (lb/hr)
1	370.3	143550	380.81	398	189.5	153495	437.66
2	402.5	142541	411.02	425	190.4	154224	469.57
3	519.2	142364	529.53	549	194.4	157464	619.31
4	516.1	145994	539.79	545	200	162000	632.51
5	449.9	147038	473.92	475	195	157950	537.49
6	343.1	143049	351.61	359	198	160380	412.48
7	310.3	143324	318.61	332	199.7	161757	384.73
8	288.5	145021	299.73	304	199.4	161514	351.75
9	277.5	144110	286.49	296	195.6	158436	335.97
10	334.8	143471	344.12	360	189.3	153333	395.45
11	341	144440	352.86	365	191.3	154953	405.18
12	343.3	143862	353.81	354	198.6	160866	407.97

Calculated Values	
di	di^2
-56.84	3231.05
-58.55	3427.80
-89.78	8060.76
-92.72	8597.01
-63.57	4041.44
-60.87	3704.86
-66.12	4372.24
-52.02	2706.37
-49.48	2448.11
-51.34	2635.36
-52.32	2737.85
-54.15	2932.31

t.025(8)	2.306
t.025(11)	2.201

For n=12	(Edi)^2	559154.71
	(SUM di^2)	48895.15
	(Edi)^2/n	46596.23
	Sd	14.46
	cc	10.06
	RM mean	386.86
	ldbarl	62.31
	RA	18.71
For n=9	(Edi)^2	249142.53
	(SUM di^2)	27865.15
	(Edi)^2/n	27682.50
	Sd	4.78
	cc	3.67
	RM mean	361.60
	ldbarl	55.46
	RA	16.35
	CEMS Avg.	417.06

Mar-01

	Field reference Method Data			CEMS Data			
Run	NO _x (ppmvd)	Q _d (dscfm)	NO _x (lb/hr)	NO _x (ppmvd)	Q _w	Q _d (dscfm)	NO _x (lb/hr)
1	425.8	138081	421.21	494	203.83	162860.2	576.36
2	426.0	135568	413.74	484	215.83	172448.2	597.94
3	425.2	141138	429.93	491	212.03	169412.0	595.91
4	413.5	137900	408.50	475	209.83	167654.2	570.51
5	414.2	141572	420.09	477	213.82	170842.2	583.81
6	416.6	140139	418.25	488	216.11	172671.9	603.67
7	417.2	141609	423.24	483	216.31	172831.7	598.03
8	422.4	141085	426.93	470	213.71	170754.3	574.94
9	409.5	139087	408.03	493	210.55	168229.5	594.16
10	421.2	139848	421.99	487	199.65	159520.4	556.55
11	447.1	139519	446.88	462	196.27	156819.7	519.04
12	434.9	138743	432.27	477	194.84	155677.2	531.98

Calculated Values	
di	di^2
-155.16	24074.05
-184.21	33932.42
-165.99	27551.21
-162.01	26246.36
-163.72	26802.93
-185.42	34380.01
-174.79	30551.67
-148.01	21906.89
-186.13	34643.57
-134.56	18105.68
-72.15	5206.22
-99.71	9942.76

t.025(8)	2.306
t.025(11)	2.201

For n=12	(Edi)^2	3355659.34
	(SUM di^2)	293343.77
	(Edi^2)/n	279638.28
	Sd	35.30
	cc	24.57
	RM mean	422.59
	ld meanl	152.65
	RA	41.94
For n=9	(Edi)^2	1628410.89
	(SUM di^2)	190387.78
	(Edi^2)/n	180934.54
	Sd	34.38
	cc	26.42
	RM mean	425.67
	ld meanl	141.79
	RA	39.52
	CEMS Avg	567.46

Jun-01

Run	Field reference Method Data			CEMS Data			
	NO _x (ppmvd)	Q _d (dscfm)	NO _x (lb/hr)	NO _x (ppmvd)	Q _w	Q _d (dscfm)	NO _x (lb/hr)
1	583	134175	560.40	586	196.27	158978.7	667.41
2	621	131420	584.67	612	181.03	146634.3	642.90
3	578	132133	547.14	584	180.57	146261.7	611.93
4	573	137925	566.18	578	175.28	141976.8	587.90
5	631	136767	618.25	631	168.77	136703.7	617.97
6	637	139383	636.07	637	168.79	136719.9	623.92
7	629	136301	614.19	630	186.32	150919.2	681.15
8	580	136691	567.97	581	181.97	147395.7	613.50
9	565	134619	544.89	590	185.32	150109.2	634.48
10	614	134484	591.55	604	183.71	148805.1	643.89

Calculated Values	
di	di^2
-107.01	11451.61
-58.23	3390.93
-64.79	4197.86
-21.72	471.68
0.29	0.08
12.15	147.69
-66.95	4482.71
-45.54	2073.46
-89.58	8025.23
-52.33	2738.82

t.025(8)	2.306
t.025(9)	2.262
t.025(11)	2.201

For n=10	(Edi)^2	243758.96
	(SUM di^2)	36980.08
	(Edi)^2/n	24375.90
	Sd	37.42
	cc	26.77
	RM mean	583.13
	ldbarl	49.37
	RA	13.06
	CEMS Avg.	632.50
For n=9	(Edi)^2	149542.54
	(SUM di^2)	25528.47
	(Edi)^2/n	16615.84
	Sd	33.38
	cc	25.66
	RM mean	585.66
	ldbarl	42.97
	RA	11.72
	CEMS Avg.	628.62